

Physics-Informed Neural Network for High-Speed Fluid Dynamics

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ABSTRACT

In this work, we demonstrate how physics-informed neural networks (PINNs) can solve various fluid dynamics problems at high-speed. The goal is to solve fundamental fluid dynamics problems by solving non-linear partial differential equations (PDEs) at high speed. We will deal with one-dimensional and two-dimensional Euler equations to solve Sod shock-tube problems, oblique shock, and expansion wave problems with vanilla PINNs and modified PINNs like weighted PINNs, and domain extension. The proposed method can learn the discontinuity present in the domain with dispersion error, but modified PINNs can minimize the error. This paper also demonstrates how the positivity condition can be enforced to solve the entropy equation across the shock. The proposed method solves the flow field for the problems mentioned above.

Keywords: Physics-Informed Neural Network, Partial Differential Equations, Euler Equations, Positivity Condition.

1 Introduction

The engineering problems are complex and can be defined by partial differential equations (PDEs). Recently, data-driven methods based on deep learning have been used to solve complex problems with the help of deep neural networks (DNNs) [1]. Such data-driven methods require large amounts of data to find generalized solutions. Finding interpretable and generalizable models in domains with limited data for complex physical systems is still tricky. This makes it necessary to understand the physics of a problem to create a generalized model by solving the PDEs. Raissi *et al.* [2] presented physics-informed neural networks (PINNs) for the first time. Such a method can solve the PDEs by the use of DNNs. It has been shown that PINNs are effective at solving forward and inverse problems associated with a variety of PDEs [3]. Since its initial publications, many contributions have been made in solving high-speed fluid dynamics problems [4, 5, 6]. Mao *et al.* [4] presented some modifications to traditional methods for solving high-speed flow problems like residual points clustering, dynamic weights, and adaptive activation functions. They have also shown the comparison of results obtained using the conventional PINNs method with those using the modifications they have made. In the paper by Jagtap *et al.* [5], a general equation for entropy conditions is given: an inequality for a non-smooth solution (shock).

In this paper, we consider the shock wave problem in supersonic compressible flows. The governing equations for shock wave problems are compressible Euler equations, which have discontinuities even though the initial states are smooth. We have shown that the traditional PINNs fail, when there is a discontinuity or shock in the domain, resulting in an inaccurate solution. However, some modifications to the PINNs approach can truncate the error in the solution. The weighted physics-informed neural networks (W-PINNs) is one of the modifications [6], where some random weights are assigned to the PDEs along with boundary and initial conditions (B.C./I.C.). First, we have solved 1D Sod shock tube problem, which deals with 1D Euler's equations where the initial condition contains the discontinuity in density and pressure. We have extended the algorithm from the 1D case of shock tube to 2D cases of

oblique shock and expansion wave problems. The positivity condition must be implemented for the 2D cases in order to obtain an accurate solution.

2 Numerical Methodology

A deep neural network is a type of neural network (NN) consisting of two or more layers of nodes in which each node is connected to all nodes in the previous and subsequent layers. Consider the input parameter $\mathbf{X}$ at the input layer and the output parameter $\mathbf{Y}$ at the output layer. When passing forward, each node transforms the input parameter $\mathbf{X}$ into the nonlinear weighted function as $f(\mathbf{W}^T \mathbf{X} + \mathbf{b})$. The weights and biases are trainable parameters, denoted by $\theta = \{\mathbf{W}, \mathbf{b}\}$. As we progress to the output layer, we can predict the solution represented by $\hat{\mathbf{Y}}$. As the output $\hat{\mathbf{Y}}$ is predicted, mean square error (MSE_{Total}) is calculated as shown in Eq. (1), where $\mathbf{Y}$ is the exact solution. N_T represents the total number of training data points respectively. Having computed the MSE_{Total} , we can use the chain rule to back-propagate and determine the gradient of the loss function with respect to the weights and biases.

$$MSE_{Loss}(\theta) = \frac{1}{N_T} \sum_{i=1}^{N_T} |\hat{\mathbf{Y}}_i - \mathbf{Y}_i|^2 \quad (1)$$

To understand the PINNs, we deal with 1D Euler's equation defined in domain $\Omega \subset \mathbb{R}$, which is given by Eq. (2)

$$\frac{\partial \mathbf{U}}{\partial t} + \mathbf{A} \frac{\partial \mathbf{U}}{\partial x} = 0, (x, t) \in \Omega \times [0, T] \quad (2)$$

where, $\mathbf{U} = (\rho, u, p)^T, \mathbf{A} = \begin{pmatrix} u & \rho & 0 \\ 0 & u & \frac{1}{\rho} \\ 0 & \rho a^2 & u \end{pmatrix}$

and $a = \sqrt{\frac{\gamma P}{\rho}}$ is the speed of sound. For Sod shock-tube problem, the initial condition is of the form mentioned in Eq. (3) with Dirichlet boundary condition.

$$\mathbf{U}(x, 0) = \mathbf{U}_0 = \begin{cases} \mathbf{u}_L & \text{if } x < x^* \\ \mathbf{u}_R & \text{if } x > x^* \end{cases} \quad (3)$$

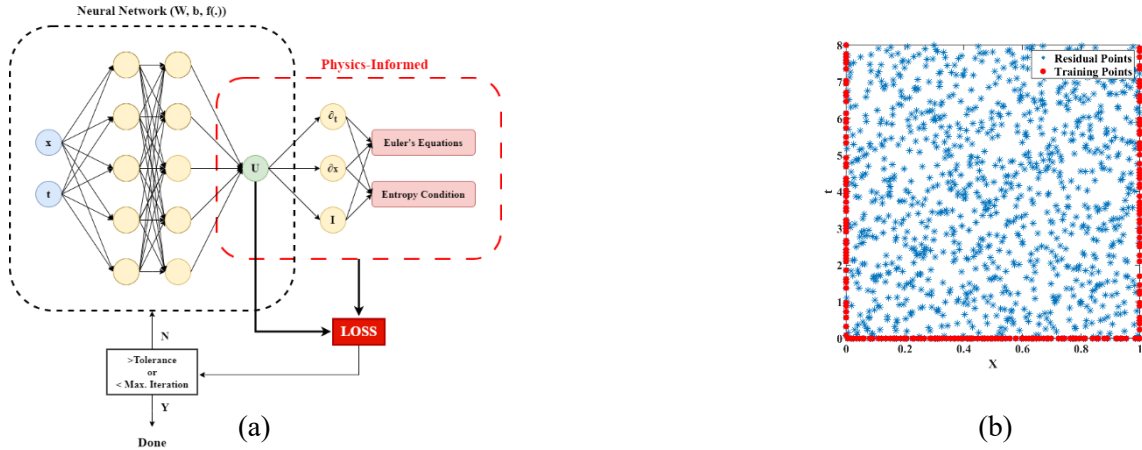


Figure 1: (a) Schematic representation of the PINNs to solve Euler equation. (b) Distribution of residual and training points.

In the traditional PINNs, we have to predict the unknown parameters generally the dependent parameter which are to be considered at the output layer. The independent parameters position and time, are considered at the input layer and the total loss function (MSE_{Total}), which is the cumulative sum of the strong form of PDEs with the B.C./I.C., as shown in Eq. (4). The trainable parameters must be trained iteratively by minimizing the total loss function (MSE_{Total}). The final predictions will be made when the total loss reaches a tolerance value or reaches the present number of iterations.

$$MSE_{Total} = MSE_{PDE} + MSE_{IC} + MSE_{BC} \quad (4)$$

$$MSE_{Total} = \frac{1}{N_T} \sum_{i=1}^{N_T} \left| \frac{\partial \hat{U}}{\partial t} + A \frac{\partial \hat{U}}{\partial x} - 0 \right|^2 + \frac{1}{N_T} \sum_{i=1}^{N_T} |\hat{U}(x, 0) - U(x, 0)|^2 + \frac{1}{N_T} \sum_{i=1}^{N_T} |\hat{U}(0, t) - U(0, t)|^2$$

An additional conservation law can be added to the governing Euler equations for smooth solutions as shown in Eq. (5.a) and an inequality for discontinuity case Eq. (5.b). Below is for formulation for 2D Euler equation case, where $\eta(U)$ and $\phi(U)$ are entropy-entropy flux pair [8, 9].

$$\eta_t + \phi_{1x} + \phi_{2y} = 0 \quad (5.a)$$

$$\eta_t + \phi_{1x} + \phi_{2y} \leq 0 \quad (5.b)$$

For the Euler equations, the most common choice of the entropy-entropy flux pair is,

$$\eta \triangleq \frac{-\rho s}{\gamma-1}, \quad \phi \triangleq \frac{\rho u s}{\gamma-1} \quad (6)$$

Where the entropy $s = \log\left(\frac{p}{\rho^\gamma}\right)$. One must pay attention that the entropy can not be negative; hence we have to enforce the positivity conditions on the density ($\hat{\rho}$) and pressure ($\hat{p}$). To model such PINNs model the output layer should be selected from $\max(\alpha, \hat{\rho})$ and $\max(\alpha, \hat{p})$. Here, parameter α is a small positive constant while helps to castigate the negative values of density and pressure. Since α is a hyperparameter, it requires thorough training. Moreover, we need to bring the equation into equality before using it as a loss function in model training. One approach is to use a constant (Λ) as demonstrate in Eq. (7) below.

$$\eta_t + \phi_{1x} + \phi_{2y} + \Lambda = 0 \quad (7)$$

3 Results and Discussion

3.1 Sod Shock-Tube

The fully connected NN is designed with four hidden layers with 80 neurons in each layer. We have used the L-BFGS optimization method with a learning rate of 0.01, number of iterations up-to 5000. Below is the loss function with predicted solution.



Figure 2: Sod Shock tube using W-PINNs: (a) Variation of loss with the number of iterations. (b) Variation of density along x direction at $t=0.2s$.

3.2 Expansion Wave

The predicted solution for expansion fan is shown in Fig. 3, where we have used PINN without using the outflow boundary conditions. Instead, we use the information of density gradient $\nabla \rho$ and a small number of pressure data points on the surface of the wedge. We have used $\tanh(\cdot)$ activation function and used L-BFGS optimization method to train the PINNs model. We have taken 1000 residual points, and 100 boundary points. To design the network, we have used six hidden layers with 60 neurons each.

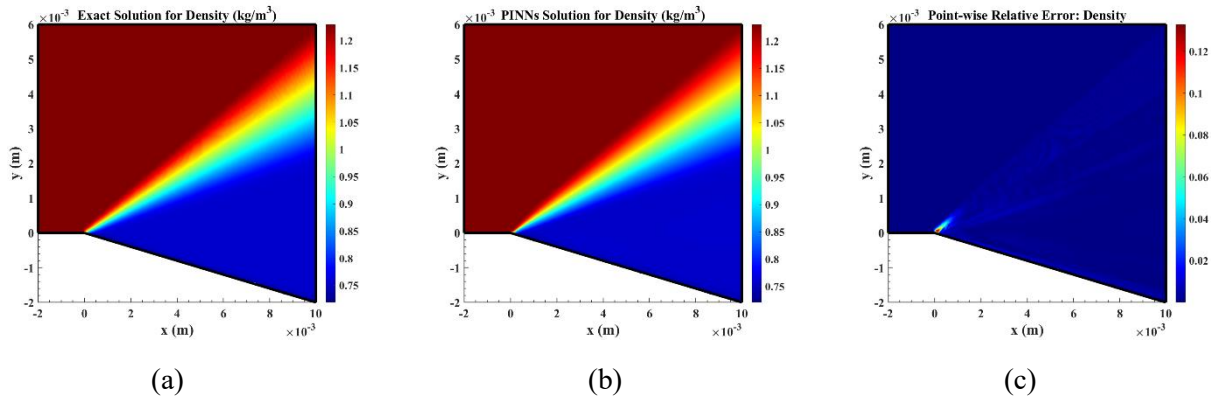


Figure 3: Expansion wave: Comparison of the density (a) Exact solution (b) PINNs solution (c) Relative point-wise error.

4 Conclusions

In this work, we solved the Sod shock-tube problem using the 1D Euler equation as a loss function with W-PINNs, a modified PINN method. At the same time, we successfully implemented the positivity condition for pressure and density to solve the entropy equation and the conservation equation by using the entropy-entropy flux pair. With this, we solved the expansion fan problem and the oblique shock problem by considering the 2D Euler equation and entropy condition and using L-BFGS optimization to train the PINN model.

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